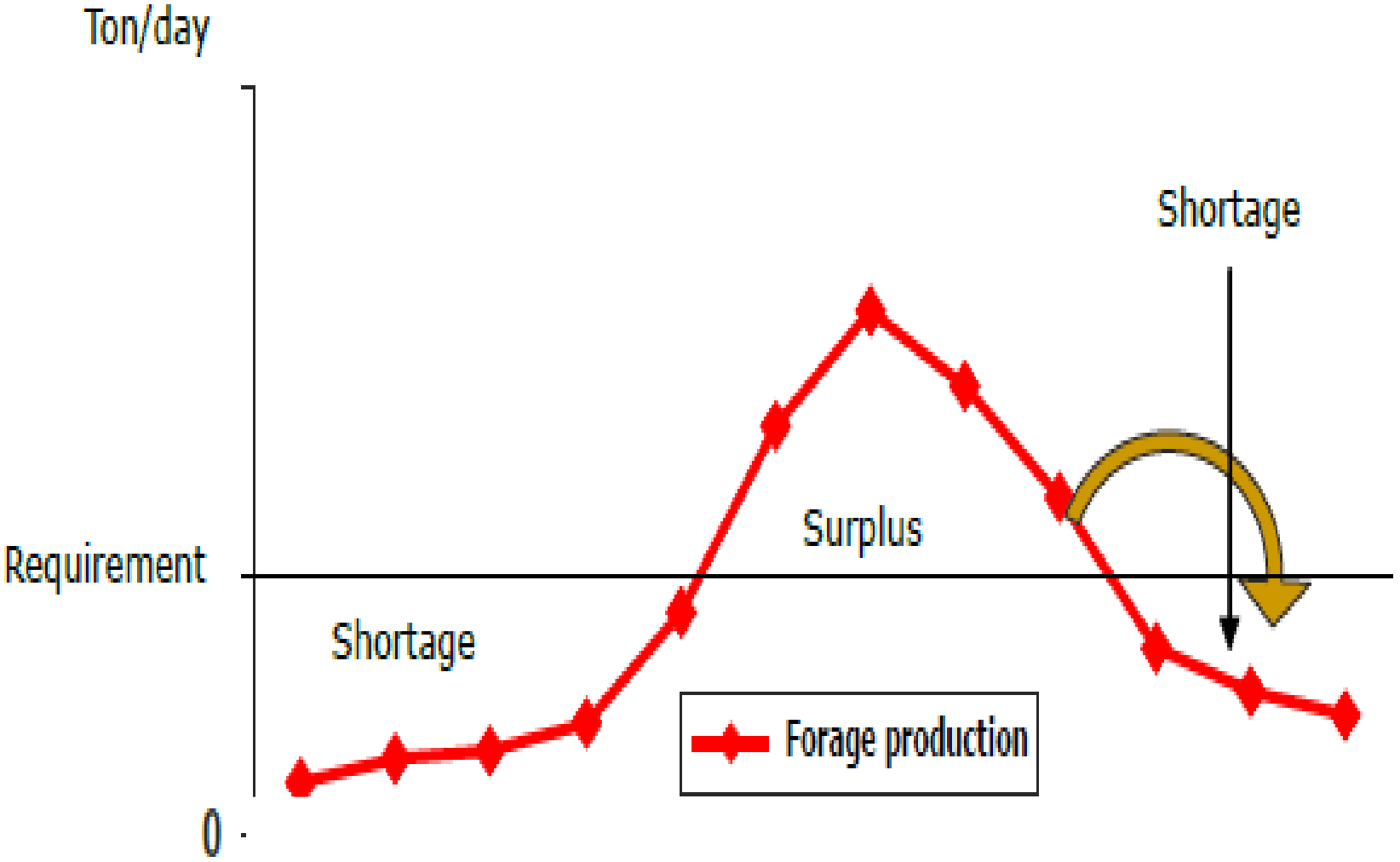
FORAGE CONSERVATION:
Haylage and Silage Production

Francisco Kanyinji
The University of Zambia,
School of Agricultural Sciences,
Department of Animal Science
Great East Road Campus
LUSAKA


Ton/day
Shortage
Requirement
Surplus
Shortage
Forage production
0

Why conserve forage?

During rainy season



- Animals are healthy,
- Gain weight,
- Reproduce, and produce a lot of milk

During dry season



- Forages dry up,
- Some get bunt by bush fires,
- The little forage available is of poor quality

As a result;

- Animals lose weight, some become sick and even die, and most of them fail to reproduce or produce.
- Also, weak or sick animals are unable to produce enough milk or meat, and cannot be used as oxen for ploughing during cultivation, or transporting things.
- Additionally when selling, they are sold at very low prices, which leads to a loss of income.

How can we counter the feed shortages during the dry seasons?



- Planting fodder crops for supplementary feeding during the dry season (e.g. sunhemp, velvet beans, cow peas, mulberry trees, moringa, etc)





- Planting improved pastures for grazing (e.g, Rhodes grass, sudan grass, napier grass).





- Cutting natural grass or improved pasture and transporting it for:



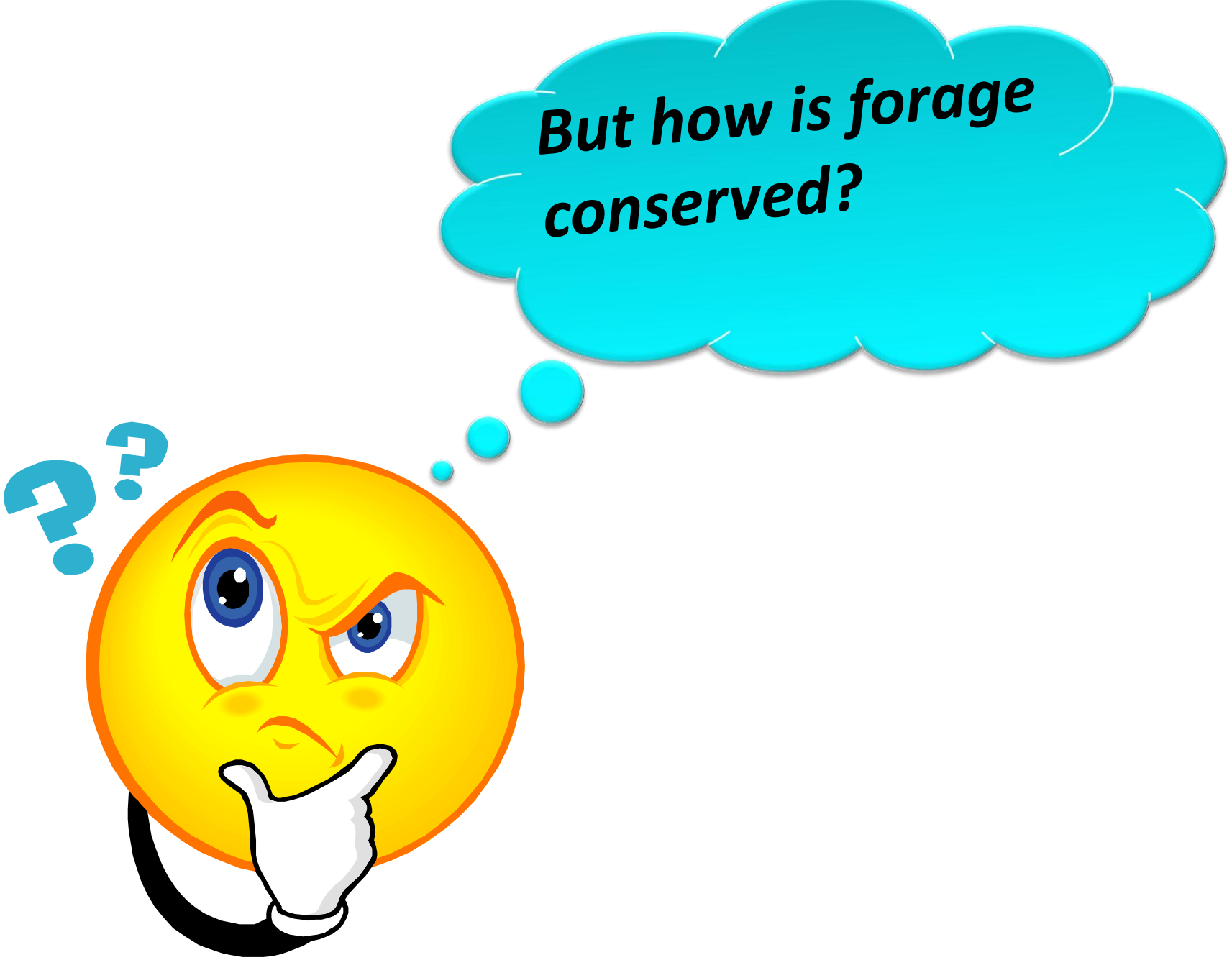


- Direct feeding to the animals if they are few in number



✓ Or making hay or silage.
This is what this section of this training is all about, ***Conservation of forages!!***




But how is forage conserved?

- Forages, whether pastures, fodder crops or crop residues can be conserved in two forms:



✓ A product of cut forage of high moisture content which has been fermented to a stable feed which resists further breakdown in anaerobic storage:



✓ this is plant or vegetative material preserved by drying for future feeding to animals. Moisture is quickly reduced from 80-10/15%.

✓ At this moisture level the grass does not rot in storage as all biological activities are arrested

However, in whatever form forage is conserved, it is important to remember that, at best, they can rarely match the nutritive value of fresh forage.

Which forage can I make hay from?



- You can also make hay from natural pasture (veldt), provided they are cut early enough (by March) to conserve the nutrients.



- However even in March, it is often too wet to dry the pasture successfully and special machinery, which is very expensive, has to be used to assist the forage to dry quickly.



Other forage crops such as maize, and forage sorghum are too thick-stemmed to dry successfully as hay. It is also difficult to make hay successfully from legumes as they drop their leaves very readily, either when handing the dried material or if allowed to get too mature before cutting. Therefore, silage is considered the better way to conserve these forage crops.

Which forage can I make silage from?















STEP 1: HARVESTING

- Harvest the material (grass, fodder crops or legumes) to be used from the field
- You can use a sickle, or simply a slasher







A rule of thumb for harvesting is, the more mature a crop is, the lower the nutrients it possesses, because it has less leafy content and more "harder" parts such as a stronger stems.

Step 2: Wilting

- Wilting is the process of making the crops loose some moisture
- Is done by spreading the harvested plants on the ground or lay them against the wall or on a rack outdoors and in the sun





A simple test; squeeze the harvested material in your hand. If there is moisture, the /hand is wet but no water dripping and it does not return to its original form quickly, then the crop is well wilted ready for silage making.

Step 3: Chopping into pieces

- Chop the wilted plants to lengths of between 1-3cm. This is because shorter chop lengths allow for better silage making.









- Take caution that you do not injure yourself

Step 4: compacting the chopped material

This stage may be done in two ways;

- You may want to mix molasses with the chopped forages as you fill and compact it in the container.





- Or you may prefer to compact it as it is without adding molasses. This is common when ensiling maize when you opt to utilize the natural sugars in the maize plant



Whatever you decide, compacting must be done well, thoroughly and layer by layer









Compact the chopped forages as you fill the plastic bag. The containers can be simple shopping plastic bags



Step 5: Closing to exclude air

- To strengthen the plastic bag, put a second layer of bag and tie it tightly. Put a third layer bag and tie it tightly as well
- These three layers of plastic bags are adequate not easily tear and allow air in.



- To avoid stretching and tearing of plastic bags when compacting, use a removable mould (e.g. half plastic drums). This makes compaction easy and protects plastic from stretching and tearing





- After compacting, tie the mouth of the bag and remove from the mould.
- Store the bags safely until feeding time sometime later




What about
hay?how is it made?

- Hay is forage dried by sun and wind, which is conserved as standing hay or cut.
- Depending on fodder crop from which the hay is made, the following are the different types of hay:



✓ **Legume hay**: This has high feeding value and is especially rich in digestible crude protein, vitamins and minerals.

✓ It is highly palatable. Examples are cowpeas, soya beans, velvet beans, and sunn hemp hay.



➢ **Non-legume hay**: This is hay from predominantly grasses and it is generally less palatable than legume hay



Steps in making hay

- You can cut the fodder with a sickle though causes some back pains because bending when doing the job



- Or specially made fodder cutter called a Scythe. This tool is especially suitable for women as it eliminates bending (as in the case of the traditional sickle).



- Handle the dried hay, for example natural grass, gently, as the leaves especially of legumes become crispy and easily break from the plant, which leads to loss of the most nutritious part of the plant.



Drying cut forages

- After cutting, dry the fodder as quickly as possible. Spread the fodder in the field to dry for about 4 hours.



- Once the top is dry, use a rake to turn the fodder and dry again for approximately 4–5 hours.



Once dried, hay can be gathered loose or can be baled for easier storage and transport. Baling is strongly encouraged as it saves storage space, avoids wastage at feeding and facilitates easy measurement for sale/marketing

Steps for bailing hay



Step 1: *Put strings in the box*



Step 2: *Fill in the dried hay and compact it*



Step 3: *Tie the strings tightly*



Step 4: *Lift the box to release the bail*

Where a wooden box is not available, a pit with the same measurements as the size of the bale desired can be made and used as a mould.

- Dig a pit on the ground measuring, for example 100 x 60 x 60cm (Length, width and height).



- Put strings in the pit for tying the hay as shown.



- Fill in as much grass as possible while compacting



- Compact the grass by trampling on with your feet.



- Tie the compressed grass in the pit tightly



- Bale of hay removed from the pit.



- The bales should be stored under a roof and stacked on raised wooden structures. The hay rack provides sufficient ventilation and it helps avoid spoilage caused by rain, sun, or insects like termites.

